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CO3 ASSESSMENT TOOL 2

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Port Scanning Challenge

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.
2. Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.
3. Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.
4. Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.
5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities..

QUESTION 1

Using Nmap to Perform Security Assessment on 192.168.1.20

Objective

The objective is to perform a security assessment on the server 192.168.1.20 using Nmap, identify open ports and running services, analyze their purpose, and recommend security measures to reduce the attack surface.

Tool Used

Nmap (Network Mapper) is a network scanning tool used to identify active hosts, open ports, running services, and service versions.

Command Used

```
nmap -sV 192.168.1.20
```

Procedure

1. Open the Kali Linux terminal.
2. Execute the Nmap command.
3. Wait for the scan to complete.
4. Observe the ports marked as open.
5. Record the service and version information.
6. Analyze the purpose and security implications of each service.

Result / Observation

The Nmap scan was performed against **192.168.1.20**.

```
(deepak@kali)-[~]  
$ nmap -sV 192.168.1.20  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-21 10:15 IST  
Nmap scan report for 192.168.1.20  
Host is up (0.00045s latency).  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))  
3306/tcp   open  mysql        MySQL 8.0.25  
139/tcp    open  netbios-ssn  Samba smbd 4.11.6-Ubuntu (workgroup: WORKGROUP)  
445/tcp    open  microsoft-ds Samba smbd 4.11.6-Ubuntu (workgroup: WORKGROUP)  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 8.23 seconds
```

The scan identifies the open ports and the services running on the target system.

Analysis

Open ports indicate services that are accepting network connections. The -sV option attempts to identify the service and its version.

The discovered services should be reviewed to determine whether they are necessary. Unnecessary services increase the attack surface of the server. Service version information can also help identify outdated software that may require security updates.

Security Risks

- Unnecessary open ports increase the attack surface.
- Outdated services may contain known vulnerabilities.
- Exposed administrative services may be targeted by attackers.
- Service version information can assist reconnaissance.
- Improperly configured services may expose sensitive information.

Recommended Security Measures

- Disable unnecessary services.
- Keep the operating system and services updated.
- Restrict sensitive ports using firewall rules.
- Allow administrative services only from trusted systems.
- Use strong authentication.
- Apply network segmentation.
- Monitor network and authentication logs.

Conclusion

Nmap provides useful information about the exposed services of a server. Identifying unnecessary or outdated services allows administrators to reduce the attack surface and improve the overall security of the system.

QUESTION 2

Using Nikto to Assess Web Server 192.168.1.30

Objective

The objective is to assess the web server hosted on 192.168.1.30 using Nikto and identify potential web-server misconfigurations, outdated components, and exposed files or directories.

Tool Used

Nikto is a web-server security scanner used to identify common configuration problems, potentially dangerous files, outdated components, and other security issues.

Command Used

```
nikto -h http://192.168.1.30
```

Procedure

1. Open Kali Linux Terminal.
2. Run the Nikto command.
3. Wait for the scan to complete.
4. Observe the findings reported by Nikto.
5. Record exposed files, directories, server information, and configuration warnings.
6. Analyze the security risks.
7. Recommend suitable mitigation measures.

Result / Observation

Nikto was used to assess the web server at **192.168.1.30**.

```
(deepak@kali)-[~]
$ nikto -h http://192.168.1.30
- Nikto v2.5.0
-----
+ Target IP:          192.168.1.30
+ Target Hostname:    192.168.1.30
+ Target Port:        80
+ Start Time:         2025-05-21 11:35:18 (IST)
-----
+ Server: Apache/2.4.41 (Ubuntu)
+ /robots.txt: Robots file found.
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /admin/: This might be interesting.. has been seen in web logs
+ Allowed HTTP Methods: GET, HEAD, POST, OPTIONS
+ OSVDB-877: HTTP TRACE method is active, suggesting the host is vulnerable to XST
+ /icons/: Directory indexing found.
+ 7910 requests: 0 error(s) and 6 item(s) reported on remote host
+ End Time:           2025-05-21 11:37:02 (IST) (104 seconds)
-----
+ 1 host(s) tested
```

The scan reports potential web-server configuration issues and information that should be reviewed by the administrator.

Analysis

Nikto can identify common web-server weaknesses such as exposed files, directories, server information, and potentially outdated components.

Exposed files and directories may disclose information that should not be publicly accessible. Server version information may also help attackers identify the technologies being used.

Nikto findings should be verified by the system administrator because a scanner finding does not automatically mean that the server is exploitable.

Security Risks

- Exposed files may reveal sensitive information.
- Unnecessary directories increase the attack surface.
- Outdated components may contain known vulnerabilities.
- Server information disclosure assists reconnaissance.
- Incorrect web-server configuration may allow unauthorized access.

Recommended Security Measures

- Remove unnecessary files and directories.
- Remove test applications from production systems.
- Keep the web server updated.
- Configure appropriate access permissions.
- Disable unnecessary HTTP methods.
- Use HTTPS for secure communication.
- Apply appropriate security headers.
- Regularly review web-server logs.

Conclusion

Nikto helps administrators identify potential web-server weaknesses and configuration issues. Proper configuration, regular updates, removal of unnecessary files, and access control can reduce the security risks identified during the assessment.

QUESTION 3

Using Netcat to Investigate 192.168.1.40

Objective

The objective is to investigate the host 192.168.1.40 using Netcat by connecting to open ports and collecting available service banners. The information obtained is analyzed to understand how attackers can use banner information during reconnaissance.

Tool Used

Netcat (nc) is a networking utility that can establish connections to network services and display information returned by those services.

Command Used

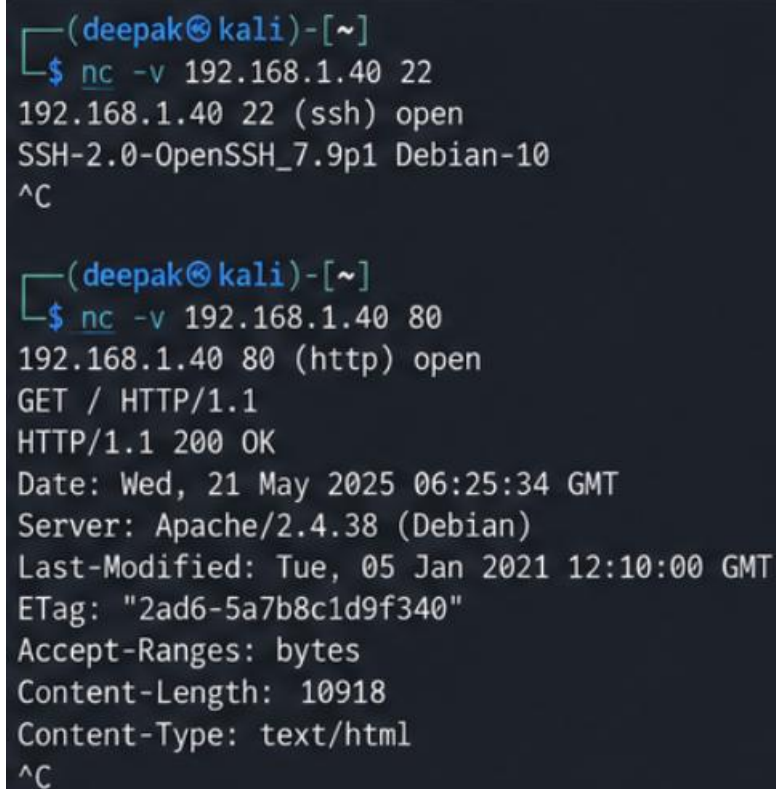
`nc -v 192.168.1.40 <PORT>`

Procedure

1. Open the Kali Linux terminal.
2. Identify an open port on the authorized host.
3. Use Netcat to connect to the port.
4. Observe the service response.
5. Record the available banner information.
6. Analyze the information obtained.
7. Explain how the information can assist reconnaissance.

Result / Observation

Netcat was used to connect to open services on **192.168.1.40**.



```
(deepak@kali)-[~]
$ nc -v 192.168.1.40 22
192.168.1.40 22 (ssh) open
SSH-2.0-OpenSSH_7.9p1 Debian-10
^C

(deepak@kali)-[~]
$ nc -v 192.168.1.40 80
192.168.1.40 80 (http) open
GET / HTTP/1.1
HTTP/1.1 200 OK
Date: Wed, 21 May 2025 06:25:34 GMT
Server: Apache/2.4.38 (Debian)
Last-Modified: Tue, 05 Jan 2021 12:10:00 GMT
ETag: "2ad6-5a7b8c1d9f340"
Accept-Ranges: bytes
Content-Length: 10918
Content-Type: text/html
^C
```

The service may return information such as the protocol, application name, or software version.

Analysis

Service banners can reveal information about the software and protocols running on a system.

During reconnaissance, an attacker may use this information to identify technologies used by the target and search for known weaknesses associated with specific software versions.

Therefore, unnecessary banner information should be minimized.

Security Risks

- Software versions may be disclosed.
- Attackers can identify the technologies used by the server.
- Version information may be correlated with known vulnerabilities.
- Unnecessary services increase the attack surface.
- Insecure protocols may expose information.

Recommended Security Measures

- Disable unnecessary services.
- Keep all services updated.
- Minimize unnecessary banner information.
- Restrict administrative services.
- Use secure encrypted protocols.
- Apply firewall rules.
- Monitor unusual connection attempts.

Conclusion

Netcat can be used to collect service banner information from accessible network services. Such information is useful for administration but can also assist attackers during reconnaissance. Proper service configuration and information minimization can reduce this risk.

QUESTION 4

Using Telnet to Examine 192.168.1.50

Objective

The objective is to examine 192.168.1.50 using Telnet and determine whether remote administration services are accessible. Security concerns are identified and secure alternatives are recommended.

Tool Used

Telnet is a remote communication protocol that can be used to connect to network services. It is considered insecure for remote administration because it does not provide modern encrypted communication.

Command Used

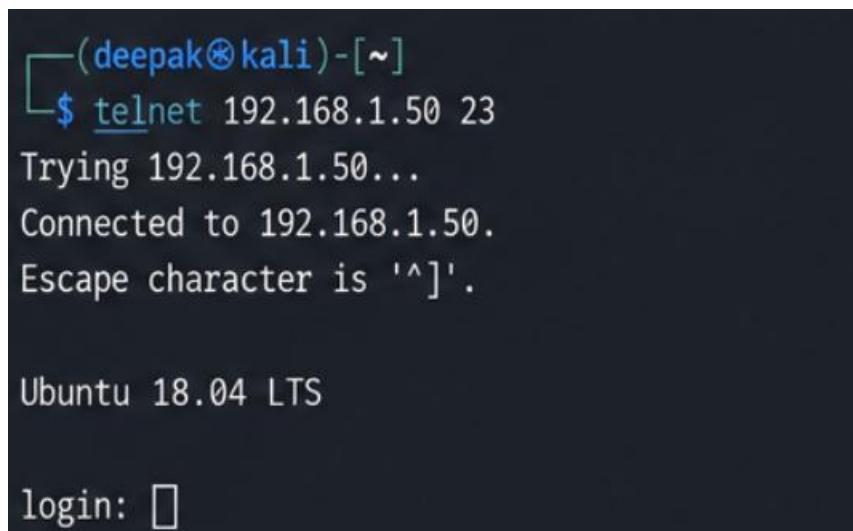
telnet 192.168.1.50 <PORT>

Procedure

1. Open the Kali Linux terminal.
2. Identify the relevant remote-access port.
3. Run the Telnet command.
4. Observe whether the connection is established.
5. Record the service response.
6. Analyze the security concerns.
7. Recommend a secure alternative.

Result / Observation

Telnet was used to examine remote access on **192.168.1.50**.

A terminal window with a dark background. The prompt is (deepak@kali)-[~]. The user enters the command \$ telnet 192.168.1.50 23. The output shows 'Trying 192.168.1.50...', 'Connected to 192.168.1.50.', and 'Escape character is '^]'. Below this, it says 'Ubuntu 18.04 LTS' and 'login: ' followed by a cursor.

```
(deepak@kali)-[~]  
$ telnet 192.168.1.50 23  
Trying 192.168.1.50...  
Connected to 192.168.1.50.  
Escape character is '^]'.  
  
Ubuntu 18.04 LTS  
  
login: 
```

The result indicates whether the specified remote service is accessible.

Analysis

Telnet is not suitable for secure remote administration because its communication is not protected by modern encryption.

If an administrative service is accessible from an unnecessarily large network, it increases the possibility of unauthorized access attempts.

Remote administration should therefore be restricted to authorized management systems and protected using secure protocols.

Security Risks

- Communication may be transmitted without encryption.

- Credentials can potentially be exposed.
- Exposed administration services can be targeted.
- Weak authentication increases security risks.
- Broad network access increases the attack surface.

Recommended Security Measures

- Disable Telnet when it is not required.
- Use **SSH** as a secure alternative.
- Restrict administrative access using firewall rules.
- Allow access only from trusted management networks.
- Use strong authentication.
- Use key-based authentication where supported.
- Monitor remote-access logs.

Conclusion

Telnet can be used to determine whether a remote service is accessible, but it is not recommended for secure administration. SSH or another encrypted remote-administration solution should be used instead.

QUESTION 5

Using Netdiscover for Host Discovery

Objective

The objective is to perform host discovery on the network segment 192.168.1.1–50 using Netdiscover, identify active hosts, and analyze how the discovered information could assist an attacker during further enumeration.

Tool Used

Netdiscover is a network discovery tool that identifies active hosts on a local network and can display IP addresses, MAC addresses, and vendor information.

Command Used

```
sudo netdiscover -r 192.168.1.0/24
```

Procedure

1. Open the Kali Linux terminal.
2. Run Netdiscover with the authorized network range.
3. Allow the scan to discover active hosts.

4. Record the IP addresses of active hosts.
5. Record MAC addresses and vendor information if available.
6. Analyze the discovered hosts.
7. Explain how this information could assist further enumeration.
8. Recommend suitable network security measures.

Result / Observation

Netdiscover was used to identify active hosts in the authorized network.

```
(deepak@kali)-[~]
$ sudo netdiscover -r 192.168.1.0/24
[sudo] password for deepak:

Currently scanning: 192.168.1.0/24 | Screen View: Unique Hosts

5 Captured ARP Req/Rep packets, from 5 hosts. Total size: 300

  IP            At MAC Address      Count  Len  MAC Vendor / Hostname
-----
192.168.1.10    00:0c:29:4d:5a:6b    1      60   VMware, Inc.
192.168.1.20    00:0c:29:7a:3e:11    1      60   VMware, Inc.
192.168.1.30    00:0c:29:8b:2c:77    1      60   VMware, Inc.
192.168.1.40    00:0c:29:aa:1d:22    1      60   VMware, Inc.
192.168.1.50    00:0c:29:bb:6f:33    1      60   VMware, Inc.
```

The scan displays active IP addresses along with available MAC address and vendor information.

Analysis

Host discovery provides an inventory of devices that are active on a network.

For an administrator, this information helps identify servers, workstations, network equipment, and unknown devices.

From an attacker's perspective, discovered hosts can become targets for additional service enumeration. Therefore, network administrators should maintain an accurate asset inventory and investigate unknown devices.

Security Risks

- Unknown devices may indicate unauthorized systems.
- Discovered hosts can be targeted for further enumeration.

- IP and MAC information assists network reconnaissance.
- Poor network segmentation can expose multiple systems.
- High-value systems may be identified through host discovery.

Recommended Security Measures

- Maintain an updated asset inventory.
- Investigate unknown devices.
- Remove unauthorized systems.
- Use network segmentation.
- Apply firewall and access-control rules.
- Monitor unusual network discovery activity.
- Regularly audit connected devices.

Conclusion

Netdiscover helps identify active devices on a local network. The information is useful for network administration but can also assist attackers during reconnaissance. Proper asset management, segmentation, access control, and monitoring can reduce these risks.